K.M.S.P. Mandal’s Sant Rawool Maharaj Mahavidhyalaya Kudal **Department of Information Technology**	Date : 16-12-25 Roll No: 62 Expt No: 02	Signature

Title : Configure AAA Authentication, configure authenticate on the Console and vty lines using local AAA and verify it.

Aim : Configure a local user account on Router and configure authenticate on the console and vty lines using local AAA Verify local AAA authentication from the Router console and the PC-A client.

A. Configuration of Local AAA Authentication

1) Enable AAA on the Router : AAA must be enabled before using local authentication.
e.g. Router(config)# aaa new-model

2) Configure Local User Account : Create a user with privilege level and password stored locally.
e.g. Router(config)# username admin privilege 15 secret admin@123

3) Configure Console Line Authentication : Apply AAA authentication method (local database) to the console.
e.g. Router(config)# line console 0
Router(config-line)# login authentication local-login
Router(config-line)# exit

Create the AAA method list:
e.g. Router(config)# aaa authentication login local-login local

4) Configure VTY Line Authentication (Telnet/SSH) : Apply the same method list to VTY lines.
e.g. Router(config)# line vty 0 4
Router(config-line)# login authentication local-login
Router(config-line)# transport input telnet ssh
Router(config-line)# exit

B. Verification of Local AAA Authentication.

1. Verification from Router Console
Steps:
Exit to the console login prompt: exit or reload.
Router will now request a username and password.

Expected Output:
Username: admin
Password: ********
After successful login, user receives privileged access : Router#
This confirms local AAA authentication is working on console.

2) Verification from PC-A (Telnet/SSH)
From PC-A Command Prompt:

If testing Telnet:
PC> telnet 192.168.1.1

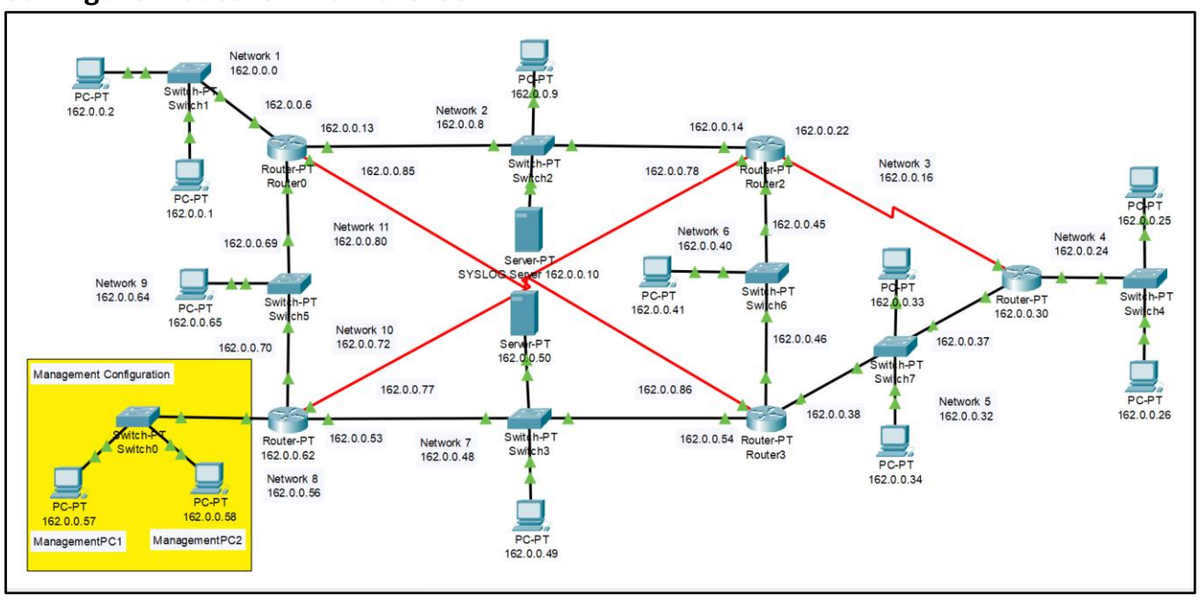
If testing SSH:
PC> ssh -l admin 192.168.1.1

Expected Output:
Trying 192.168.1.1 ... Open
Username: admin
Password: ********
Router#

Successful login confirms that VTY AAA authentication via local database is working.

ISO privilege levels (0-15)
Privilege Level 0 – Very limited access (only: exit, help, logout, enable)
Privilege Level 1 – User EXEC mode (basic monitoring, no configuration)
Privilege Level 2 – Custom level (no default commands, admin assigns)
Privilege Level 3 – Custom level (admin assigns commands)
Privilege Level 4 – Custom level (admin assigns commands)
Privilege Level 5 – Custom level (often used for junior admins)
Privilege Level 6 – Custom level (often used for technicians)
Privilege Level 7 – Custom level (troubleshooting access if assigned)
Privilege Level 8 – Custom level (NOC / support use if assigned)
Privilege Level 9 – Custom level (senior technician)
Privilege Level 10 – Custom level (senior engineer)
Privilege Level 11 – Custom level (internal admin roles)
Privilege Level 12 – Custom level (high-level support)
Privilege Level 13 – Custom level (lead engineer)
Privilege Level 14 – Custom level (near full admin, supervisor)
Privilege Level 15 – Full privileged EXEC mode (complete administrative access)

Configure Routers with the ssh :



Configure ssh Routers -> Router1

```
YashodipRouter1>enable
Password:
YashodipRouter1#config terminal.
YashodipRouter1(config)#ip domain-name yashinfotech.com
YashodipRouter1(config)#username admin privilege 15 password admin@123
YashodipRouter1(config)#username yashodip privilege 15 password yashodip@123
YashodipRouter1(config)#crypto key generate rsa
How many bits in the modulus [512]: 2048
YashodipRouter1(config)#ip ssh version 2
YashodipRouter1(config)#line vty 0 4
YashodipRouter1(config-line)#login local
YashodipRouter1(config-line)#transport input ssh
YashodipRouter1(config-line)#exit
```

Router2

```
YashodipRouter2>enable
Password:
YashodipRouter2#config terminal
YashodipRouter2(config)#ip domain-name yashinfotech.com
YashodipRouter2(config)#username admin privilege 15 password admin@123
YashodipRouter2(config)#username vaibhavi privilege 15 password vaibhavi@123
YashodipRouter2(config)#crypto key generate rsa
The name for the keys will be: YashodipRouter2.yashinfotech.com
How many bits in the modulus [512]: 2048
YashodipRouter2(config)#ip ssh version 2
YashodipRouter2(config)#line vty 0 4
YashodipRouter2(config-line)#login local
YashodipRouter2(config-line)#transport input ssh
YashodipRouter2(config-line)#exit
```

Router3

```
YashodipRouter3>enable
Password:
YashodipRouter3#config terminal
YashodipRouter3(config)#ip domain-name yashinfotech.com
YashodipRouter3(config)#username admin privilege 15 password admin@123
YashodipRouter3(config)#username lajari privilege 15 password lajari@123
YashodipRouter3(config)#crypto key generate rsa
How many bits in the modulus [512]: 2048
YashodipRouter3(config)#ip ssh version 2
YashodipRouter3(config)#line vty 0 4
YashodipRouter3(config-line)#login local
YashodipRouter3(config-line)#transport input ssh
YashodipRouter3(config-line)#exit
```

Router4

```
YashodipRouter4>enable
Password:
YashodipRouter4#config terminal
YashodipRouter4(config)#ip domain-name yashinfotech.com
YashodipRouter4(config)#username admin privilege 15 password admin@123
YashodipRouter4(config)#username shruti privilege 15 password shruti@123
YashodipRouter4(config)#crypto key generate rsa
How many bits in the modulus [512]: 2048
YashodipRouter4(config)#ip ssh version 2
YashodipRouter4(config)#line vty 0 4
YashodipRouter4(config-line)#login local
YashodipRouter4(config-line)#transport input ssh
YashodipRouter4(config-line)#exit
```

Router5

```
YashodipRouter5>enable
Password:
YashodipRouter5#config terminal
YashodipRouter5(config)#ip domain-name yashinfotech.com
YashodipRouter5(config)#username admin privilege 15 password admin@123
YashodipRouter5(config)#username sonia privilege 15 password sonia@123
YashodipRouter5(config)#crypto key generate rsa
How many bits in the modulus [512]: 2048
YashodipRouter5(config)#ip ssh version 2
YashodipRouter5(config)#line vty 0 4
YashodipRouter5(config-line)#login local
YashodipRouter5(config-line)#transport input ssh
YashodipRouter5(config-line)#exit
```

Router	User1	Password1	User2	Password2
R1	admin	admin@123	yashodip	yashodip@123
R2	admin	admin@123	vaibhavi	vaibhavi@123
R3	admin	admin@123	lajari	lajari@123
R4	admin	admin@123	shruti	shruti@123
R5	admin	admin@123	sonia	sonia@123

Accessing the ssh using PC ->

```
Packet Tracer PC Command Line 1.0
C:\>ssh -l admin 162.0.0.30
Password:
YashodipRouter3#show clock
*1:12:26.673 UTC Mon Mar 1 1993
host not responding
```

local AAA authentication using the line console

```
YashodipRouter1>enable
Password:
YashodipRouter1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
YashodipRouter1(config)#aaa new-model
YashodipRouter1(config)#aaa authentication login CONSOLE local
YashodipRouter1(config)#line console 0
YashodipRouter1(config-line)#login authentication CONSOLE
YashodipRouter1(config-line)#exit

User Access Verification
Username: admin
Password:
YashodipRouter1>show clock
*16:44:4.268 IST Wed Apr 9 2025
YashodipRouter1>logout
```

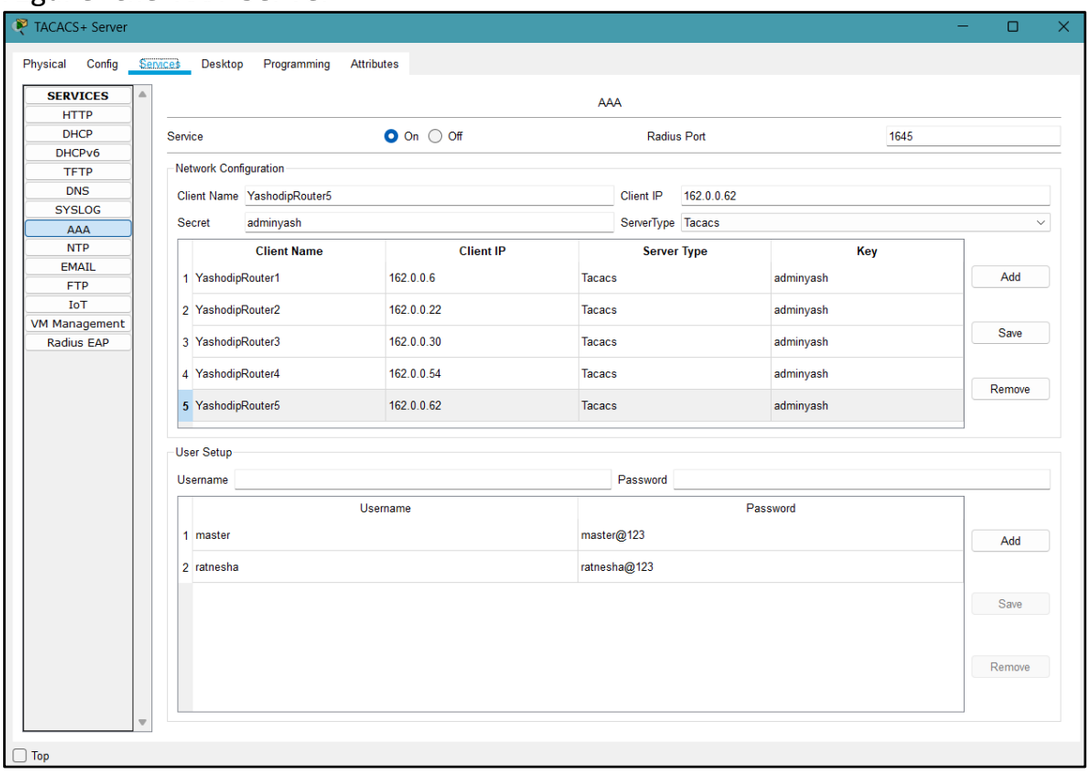
local AAA authentication using the line vty

```
YashodipRouter1>enable
Password:
YashodipRouter1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
YashodipRouter1(config)#aaa new-mode
YashodipRouter1(config)#aaa authentication login VTY local
YashodipRouter1(config)#line vty 0 4
YashodipRouter1(config-line)#login authentication VTY
YashodipRouter1(config-line)#exit
YashodipRouter1(config)#exit

User Access Verification
Username: yashodip
Password:
YashodipRouter1>show clock
*16:44:24.798 IST Wed Apr 9 2025
YashodipRouter1>logout
```

Centralized AAA authentication using the TACACS+ :

Configure the AAA server



Configure the Router ->

```
YashodipRouter4>enable
Password:
YashodipRouter4#config terminal
YashodipRouter4(config)#aaa new-model
YashodipRouter4(config)#tacacs-server host 162.0.0.50
YashodipRouter4(config)#tacacs-server key cisco
YashodipRouter4(config)#aaa authentication login default group tacacs local
YashodipRouter4(config)#line vty 0 4
YashodipRouter4(config-line)#login authentication default
YashodipRouter4(config-line)#exit
YashodipRouter4(config)#line console 0
YashodipRouter4(config-line)#login authentication default
YashodipRouter4(config-line)#exit
YashodipRouter4(config)#exit
*Mar 01, 00:07:08.077: SYS-5-CONFIG_I: Configured from console by console
YashodipRouter4 con0 is now available
```

AAA tacacs+ authentication from router

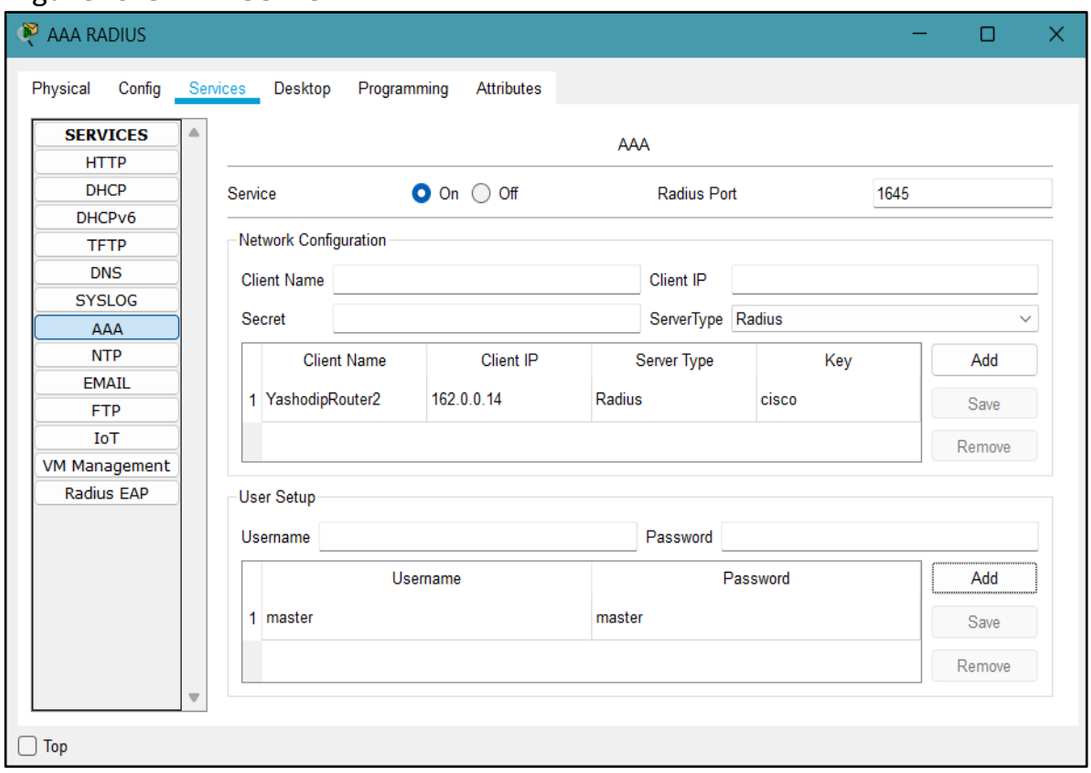
```
User Access Verification
Username: super
Password:
YashodipRouter4>show clock
*0:8:25.833 UTC Mon Mar 1 1993
YashodipRouter4>logout
```

AAA tacacs+ authentication from PC(device)

```
Packet Tracer PC Command Line 1.0
C:\>ssh -l super 162.0.0.54
Password:
YashodipRouter4>show clock
*0:21:42.705 UTC Mon Mar 1 1993
YashodipRouter4>logout
```

Centralized AAA authentication using the RADIUS :

Configure the AAA server



Configure the router for the Radius server ->

```
YashodipRouter2>enable
Password:
YashodipRouter2#config terminal
YashodipRouter2(config)#aaa new-model
YashodipRouter2(config)#aaa authentication login default group radius local
YashodipRouter2(config)#radius-server host 162.0.0.10
YashodipRouter2(config)#radius-server key cisco
YashodipRouter2(config)#line vty 0 4
YashodipRouter2(config-line)#login authentication default
YashodipRouter2(config-line)#exit
YashodipRouter2(config)#line console 0
YashodipRouter2(config-line)#login authentication default
YashodipRouter2(config-line)#exit
YashodipRouter2(config)#exit
*Mar 01, 00:34:46.3434: SYS-5-CONFIG_I: Configured from console by console
```

AAA radius+ authentication from router

```
User Access Verification
Username: master
Password:
YashodipRouter2>show clock
*0:35:1.832 UTC Mon Mar 1 1993
YashodipRouter2>logout
```

AAA radius+ authentication from PC(device)

```
Packet Tracer PC Command Line 1.0
C:\>ssh -l master 162.0.0.14
Password:
YashodipRouter2>show clock
*0:37:54.599 UTC Mon Mar 1 1993
YashodipRouter2>logout
```